

Fundamentals of Software Design

The two forces behind every good design — and a law that keeps them honest.

From 'What' to 'How'

Last time: design is decision-making, and it lives on a continuum with architecture. Today we get concrete tools to judge those decisions.

Cohesion

Does each part do one thing well?

Coupling

How tangled are the parts with each other?

Change

Will the next requirement be cheap — or painful?

Learning Objectives

- Define cohesion and coupling — and recognise them in real code.
- Explain why 'high cohesion, low coupling' is the golden rule of design.
- See how coupling turns a small change into an expensive one.
- Apply the Law of Demeter to tame dependency chains.

SECTION 01

Cohesion

How well the things inside a module belong together.

Two Forces Shape Every Design

Almost everything we call 'good design' comes down to balancing two measurable forces. Keep them in mind for the rest of the bootcamp:

Cohesion (want HIGH)

- How focused a single module is
- One clear responsibility
- Measured within a module

Coupling (want LOW)

- How dependent modules are on each other
- How much one must know about another
- Measured between modules

What Is Cohesion?

Cohesion is the degree to which the elements inside a module belong together — how single-minded that module is about its job.

High cohesion = a class or function that does **one thing**, and everything in it serves that thing.

Low vs High Cohesion

A 'God' utility that does unrelated jobs vs. focused classes with one responsibility each:

Java*One class, four reasons to change*

```
// LOW cohesion – unrelated responsibilities in one class
class Utils {
    void saveUser(User u) { /* database */ }
    void sendEmail(String to) { /* SMTP */ }
    double calcTax(Order o) { /* finance */ }
    String formatDate(Date d) { /* formatting */ }
}
```

Split into **UserRepository**, **EmailService**, **TaxCalculator** — each cohesive, each with a single reason to change.

A Password Value Object

The opposite of the god-utility. One concept, one class: validation, random creation, comparison, formatting and masking for a password all live here — and nothing unrelated does.

Java*every method is about one thing — a Password*

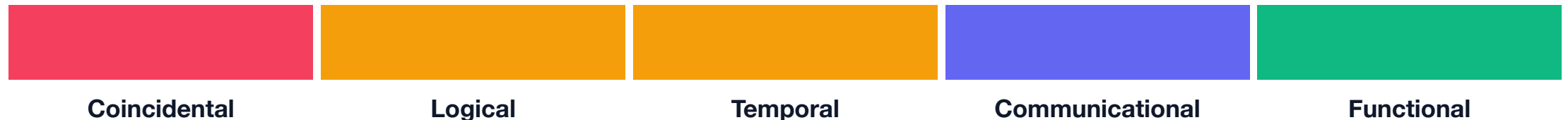
```
public final class Password {    // one concept, one class
    // create & validate — one place owns the rules
    public static Password of(String raw)        { ... }
    public static boolean  isValid(String raw)    { ... }
    // generate a strong random password
    public static Password random(int length)    { ... }
    // compare safely — constant-time, no early exit
    public boolean matches(String candidate)      { ... }
    // format & mask — never leak the value
    public String masked()                       { ... }
    public String maskedShowingLast(int shown)   { ... }
}
```

Cohesion Comes in Degrees

Not all cohesion is equal. Aim for the right end of this scale:

weakest

strongest



Functional cohesion — every element contributes to a single, well-defined task — is the goal. You don't need the jargon; you need the instinct.

Full taxonomy → [en.wikipedia.org/wiki/Cohesion_\(computer_science\)](https://en.wikipedia.org/wiki/Cohesion_(computer_science))

Cohesion Degrees — the Weaker End

The weak degrees bundle code for shallow reasons: accident, category, or timing. The // comment names the smell.

Coincidental

```
class Utils {  
    int add(int a, int b){}  
    void log(String message){}  
    boolean valid>Email email){}  
}  
// unrelated jobs bundled
```

Logical

```
String read(int source){  
    if(source==1) file();  
    if(source==2) net();  
    ...  
}  
// same verb, a flag picks
```

Temporal

```
void startup(){  
    loadConfig();  
    openLog();  
    connectDb();  
}  
// grouped by "when"
```

Rule of thumb: if the only thing the members share is a class name or a moment in time, cohesion is weak.

Cohesion Degrees — the Stronger End

The strong degrees bind code by real, single-minded purpose: the same data, or one well-defined task. This is the goal.

Communicational

```
Stats of(Order order){  
    var items = order.items();  
    return new Stats(  
        sum(items),  
        items.size());  
}  
// all share one data set
```

Functional

```
double tax(double amount){  
    return amount * RATE;  
}  
// one job, done well
```

Aim for functional cohesion: every element earns its place by serving the one job the module exists to do.

SECTION 02

Coupling

How much modules depend on — and know about — each other.

DEFINITION

What Is Coupling?

Coupling is the strength of the dependencies between modules — how much one module must know about another to work.

Loose coupling = modules interact through small, stable interfaces and know as **little** about each other as possible.

Tight vs Loose Coupling

Depending on a concrete class welds you to it. Depending on an interface lets you swap the other side freely:

Java — tight

```
class Report {  
    MySqlDb db =  
        new MySqlDb();  
    // welded to MySQL  
    void run() {  
        db.query(...);  
    }  
}
```

Java — loose

```
class Report {  
    final Database db;  
    Report(Database db){  
        this.db = db;    // any DB  
    }  
    void run() {  
        db.query(...);  
    }  
}
```

The loose version depends on an abstraction — swap MySQL for Postgres or a fake in tests, no change to Report.

FROM STRUCTURED DESIGN

Coupling Comes in Degrees

These classic coupling types come from Structured Design (Yourdon & Constantine) — a scale from welded-together to barely acquainted:

tightest — worst

loosest — best



Aim for data coupling — modules pass only the simple values they need. Message coupling (talk through interfaces or events) is looser still.

Full taxonomy → [en.wikipedia.org/wiki/Coupling_\(computer_programming\)](https://en.wikipedia.org/wiki/Coupling_(computer_programming))

Coupling Degrees — the Tighter End

Tight coupling reaches across boundaries: into another module's internals, its globals, or its control flow.

Content

```
class Report {  
    void hack(Engine engine){  
        engine.data[0] = 42;  
    }  
}  
// writes Engine's field
```

Common

```
static int mode;  
void writer(){ mode=1; }  
void reader(){  
    if(mode==1){ ... }  
}  
// shared global state
```

Control

```
void draw(Doc doc,  
          boolean asHtml){  
    if(asHtml){}  
    else {}  
}  
// caller drives branch
```

Smell: one module can break another by touching data or logic it was never meant to see.

Coupling Degrees — the Looser End

Loose coupling narrows the contract to exactly what is needed — ideally a few simple values.

Stamp

```
double ship(Customer customer){  
    return rate(customer.zip());  
}  
// whole object, one field used
```

Data

```
double ship(String zip){  
    return rate(zip);  
}  
// pass only what's used
```

Aim for data coupling: the caller hands over the minimum, so either side can change without disturbing the other.

Coupling in OOP: Subtyping & Message

Objects add two more types beyond Structured Design's. Inheritance binds a subclass tightly to its parent; messages bind the loosest of all.

Subtyping — tight

```
class SalesReport extends Report {  
  void run() {  
    prepare();           // parent hook  
    rows.add(record);    // parent field  
  }  
}  
// bound to Report's shape & internals
```

Message — loosest (best)

```
class Orders {  
  final Mailer mailer; // an interface  
  void place(Order order) {  
    mailer.send(order.receipt());  
  }  
}  
// only a message crosses — nothing else
```

Program to an interface, not an implementation. — *Gang of Four, Design Patterns (1994)*

Depend on abstractions, not on concretions. — *Robert C. Martin, [Dependency Inversion Principle](#) (1996)*

Semantic Coupling

The most dangerous coupling isn't in the interface. Semantic coupling is when one module leans on another's inner behaviour and assumptions — not its syntax. The compiler can't see it.

Call-order

```
parser.initialize();
parser.parse();
// parse() works only
// if initialize() ran
// – nothing enforces it
```

Magic flag

```
save(order, true);
// 'true' = skip the
// validation step;
// caller assumes what
// save() does inside
```

Partial init

```
var user = new User();
user.setName(name);
mailer.send(user);
// ok only while send()
// reads just .name –
// add a field, it breaks
```

Nothing in the syntax reveals it — so it breaks at runtime, far from the change that caused it.

 **Code Complete (Steve McConnell)** — *names semantic coupling the most insidious kind — it survives the compiler and breaks silently when the other module's internals change.*

Three Dimensions of Coupling

You cannot get rid of coupling — a system with none does nothing. The goal is not to remove it but to **balance** it. Khononov models its cost as three dimensions:

Integration Strength

How much knowledge two parts share.

Contract → **Intrusive**

e.g. a shared API contract (weak) vs. reading another module's private DB tables (intrusive).

Distance

How far apart the parts live.

Same method → **Separate systems**

e.g. two methods in one class (near) vs. two services owned by different teams (far).

Volatility

How often the parts change.

Stable → **Volatile**

e.g. the JDK's String, rock-stable, vs. a pricing rule that changes every week (volatile).

📖 **Balancing Coupling in Software Design (Khononov)** — *coupling hurts most when strength, distance and volatility are all high at once — the design job is to balance the three.*

Coupling Is a Change Multiplier

The cost of coupling shows up the day you change something. Tightly coupled code spreads one change across many files:

Tightly coupled

- One change ripples outward
- Hard to test a part in isolation
- Fear of touching anything

Loosely coupled

- Changes stay local
- Parts tested independently
- Confidence to evolve

Leaky Abstractions

Low cohesion and high coupling are one failure seen twice. A class that does not own its behavior leaks its internals — and whoever grabs those internals is now welded to them.

Leaky — internals exposed

```
class X {
  private List<Item> items;
  List<Item> getItems(){ // leaks it
    return items;
  }
}

x.getItems().add(item); // Y reaches in
x.getItems().remove(item); // Z reaches in
```

Sealed — behavior only

```
class X {
  private List<Item> items;
  void addItem(Item i){ items.add(i); }
  void removeItem(Item i){ items.remove(i); }
}

x.addItem(item); // Y asks X
x.removeItem(item); // Z asks X
```

Low cohesion IS high coupling seen from the other side — own your behavior, and expose only the interface, never the representation.

📖 **The Law of Leaky Abstractions (Joel Spolsky, 2002)** — *every abstraction leaks something — the skill is leaking only the interface, never the internals.*

High Cohesion, Low Coupling

Group what changes together. Separate what changes apart.

High cohesion pulls together

- Related behaviour lives in one place
- Easy to find and understand

Low coupling keeps apart

- Unrelated parts stay independent
- Easy to change one without the others

SECTION 03

Designing for Change

Cohesion and coupling only matter because software changes.

We Optimize for Change

You will never predict every future requirement. So you don't design to predict change — you design to absorb it cheaply when it comes.

- Isolate what varies — put likely changes behind a boundary.
- Hide decisions — callers shouldn't depend on how something works.
- Depend on abstractions, not details (we'll formalize this in SOLID).
- Small, cohesive, loosely coupled parts = a system that bends, not breaks.

Encapsulation & Information Hiding

Encapsulation is how we buy low coupling: expose a small, stable surface; hide the volatile details behind it.

Expose (stable)

- What the module does
- A small, intention-revealing API
- Types the caller truly needs

Hide (volatile)

- How it does it — algorithms, data
- Internal fields & helper types
- Third-party libraries & storage

SECTION 04

The Law of Demeter

A simple rule to stop coupling from creeping back in.

Don't Talk to Strangers

The Law of Demeter (the 'principle of least knowledge') says a method should only talk to its immediate friends — not reach through them to strangers.

- Only call methods on: itself, its fields, its parameters, objects it creates.
- Avoid long chains: `a.getB().getC().doSomething()` reaches through others.
- Each dot past the first is a new thing you've secretly coupled to.

The 'Train Wreck'

This line knows the shape of Order, Customer AND Address. Change any one and it may break:

Java

```
String city = order
    .getCustomer()
    .getAddress()
    .getCity();
// reaches 3 classes
```

C#

```
var city = order
    .Customer
    .Address
    .City;
// reaches 3 classes
```

Python

```
city = (order
    .customer
    .address
    .city)
# reaches 3 classes
```

Every getter in the chain is a dependency in disguise.

Tell, Don't Ask

Ask the object you know to do the job — let it deal with its own internals:

Java*The chain is gone — Order hides its structure*

```
// Instead of reaching through order for the city...
String city = order.shippingCity(); // ask the object you have

class Order {
    String shippingCity() {
        return customer.shippingCity(); // each object knows its own parts
    }
}
```

Now callers depend only on Order. Its internal shape is free to change.

Powerful — but Not Absolute



It buys you

- Lower coupling between classes
- Freedom to change internals
- Code that reads as intent, not plumbing



Keep perspective

- It's a heuristic, not a hard law
- Fluent builders & streams are fine
- Aim for less knowledge, not zero dots

REMEMBER THESE

Key Takeaways

- ✓ Cohesion (high) and coupling (low) are the two forces of good design.
- ✓ Cohesion: one module, one job. Coupling: know as little as possible.
- ✓ Coupling is a change multiplier — it turns small edits into big ones.
- ✓ Encapsulation buys low coupling: expose a stable surface, hide the details.
- ✓ Law of Demeter: talk to friends, not strangers — kill the train wrecks.

What's Next

MODULE 3 · SATURDAY 18 JULY

Clean Code & SOLID

with Akin Kaldıroğlu

Turning these forces into everyday habits: a Clean Code framework, then the five SOLID principles that make high cohesion and low coupling systematic.

Find These Smells for Real

Every force from this session lives in the ACME Orders codebase. Measure the cohesion and coupling, then improve them.

Refactoring course · Cohesion & Coupling

Repo github.com/kaldiroglu/refactoring-course-student-materials

IN THIS SESSION, LOOK AT

- ▶ Exercises/Ex01-CohesionCoupling.md
- ▶ .../service/OrderService.java (coupled god-service)
- ▶ .../util/Util.java (low cohesion)

Questions?

High cohesion, low coupling — everything else is detail.